Upgrading a R513A C-series Heat Pump with flash media

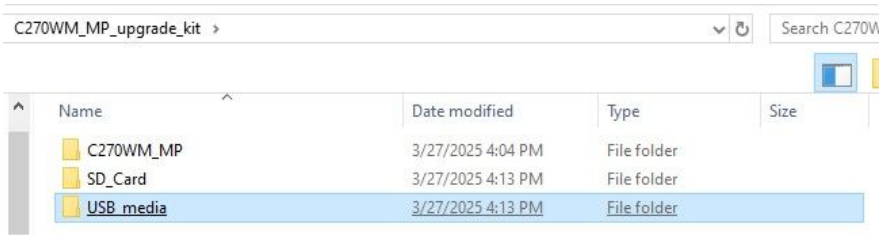
This upgrade process assumes your are upgrading an R513A C-series Heat Pump System HMI and PLC.
This document is relevant to all R513A C-series single pass, multi pass units, water or air source.

We recommend backing up the entirety of your existing PLC SD card including all files in the FCDATA01 folder prior to performing this upgrade.

Before you begin your software update, please navigate to the “Configuration” page of the Main Control Panel and record all settings. After the update you will need to confirm all settings, and some upgrades may require setting new configuration items that did not previously exist.

To begin, download the R513A upgrade zip file package that matches your specific unit model, and extract the contents.

Below is an Example of a C270 Water, Multi Pass upgrade kit file structure.
In this guide, the SD_Card Folder contents are copied to an SD Card.
The USB_media folder contents are copied to a USB media stick.
The contents of C270WM_MP will be copied to a blank SD Card after the HMI and PLC are upgraded.

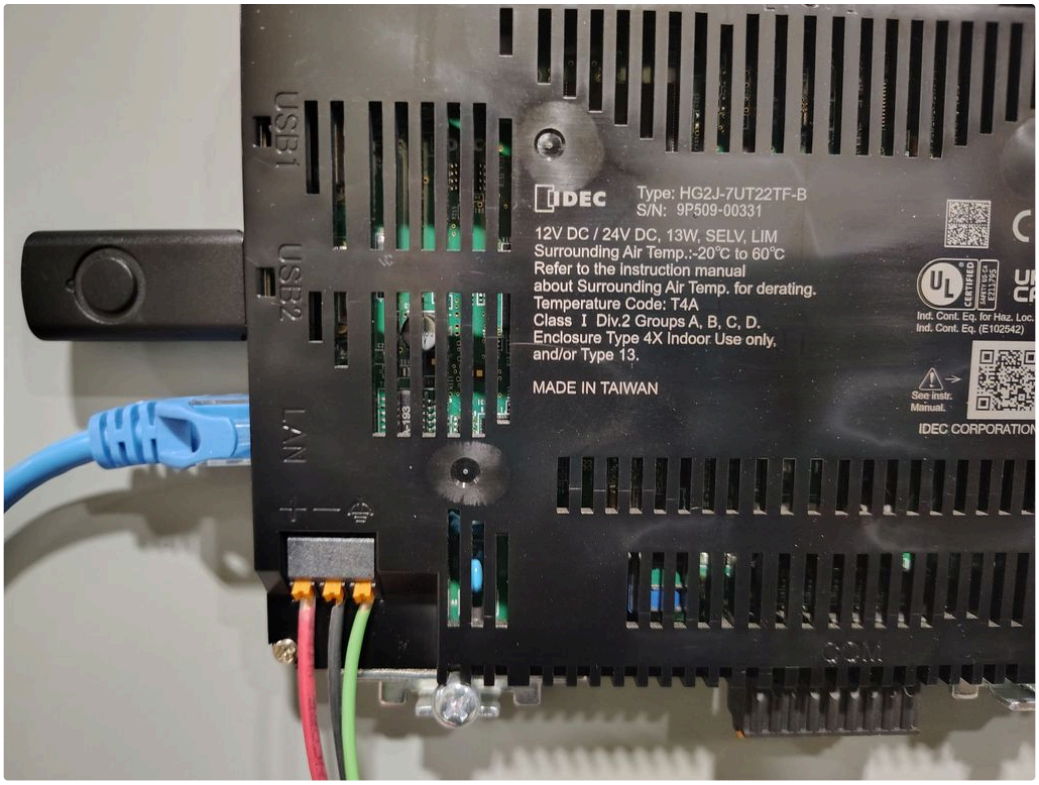


Example Flash Media upgrade kit file layout

Move the contents of the USB_Media folder to a FAT32 formatted USB-A flash drive. You should have an hgauto.ini file, and a file named R513a1308.ZNX. Be sure to place these files directly in the USB media root directory with no additional folders. Safely remove the USB device from your computer.

Next with an otherwise empty SD Card FAT32 formatted, copy the contents of the folder named SD_Card to the SD Card. You will now have an SD Card with the folder FCDATA01 in the root directory.

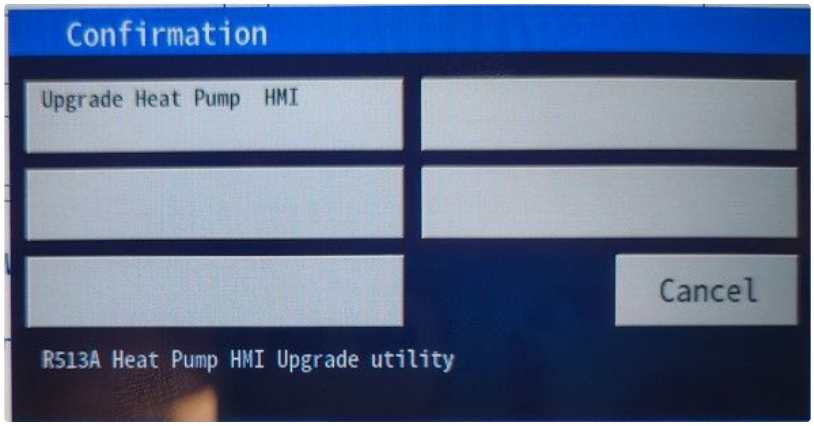
With the Heat Pump System powered on, but the system set to “system off”, open the electrical enclosure, identify the back of the screen (HMI), and insert the USB media into the USB2 slot of the Heat Pump HMI.



Insert USB Upgrade Media into USB2 as shown

USB2 is the port located immediately next to the Ethernet port. This view is the rear of the HMI which you will see after you open the Heat Pump enclosure door.

Close the enclosure door of the Heat Pump, and secure as necessary. You should now be viewing the HMI control screen. Within 10 seconds a “Confirmation” window should appear as shown .

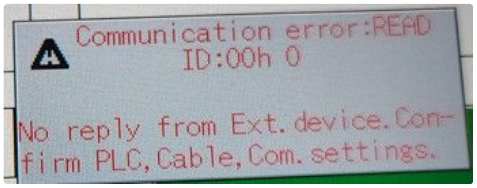


This is the Heat Pump Upgrade Utility interface.

From the Confirmation window, select **upgrade Heat Pump HMI**. This will upgrade the Heat Pump HMI.
You may be prompted for a password. The user is User and the password is 1000. The HMI may flash, go blank, and reset during the upgrade process this is normal. When completed, a confirmation window will appear. Press **Ack** to continue.

Be sure you have recorded your previous settings and configurations, and backed up the SD Card which came with the Heat Pump. Next, power off your heat pump, and remove the USB media from the HMI.

Move the contents of the SD_Card folder to an otherwise blank SD Card, you should now have an SD Card with FCDATA01 on it. Safely remove the SD Card from your computer. Insert the SD card into the PLC. Power on the PLC and HMI. The heat pump HMI and PLC may take longer than normal to initialize, while the PLC software upgrade is being installed. The PLC will click, and the HMI may display communication errors until the PLC has competed upgrading.



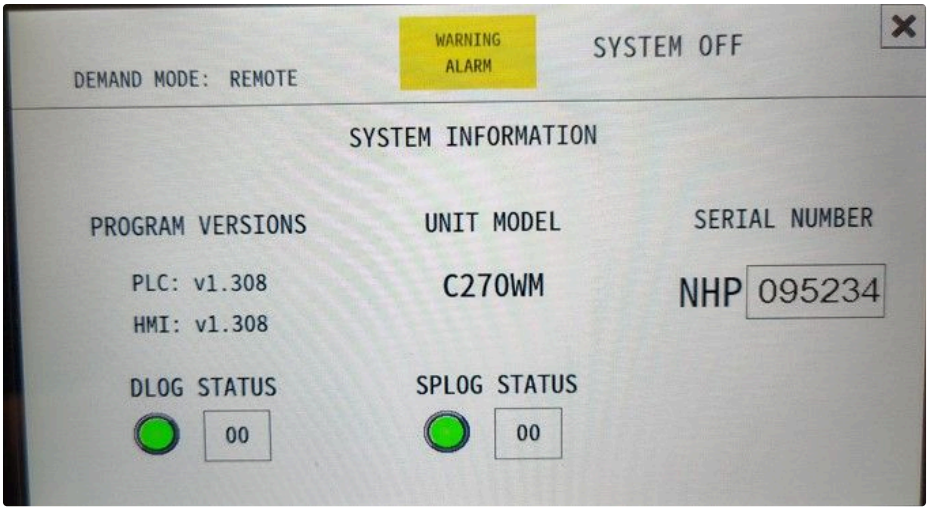
Please be patient, the upgrade process may take up to 5 minutes to complete. Once the PLC and HMI have fully initialized, communication errors will disappear.

Verify the upgrade was performed successfully. Navigate to the **SYSTEM INFO** page by holding the time display located at the top right corner of the HMI for approximately 5 seconds. Tap the **SYSTEM INFO** button located on the top left of the HMI screen. Your heat pump and HMI Version should now match.

Remove the SD Card upgrade media from the PLC.

Finally you must provision an SD Card with configuration specific to your Model Heat Pump.
From the Flash Media upgrade kit files downloaded and extracted, a folder with the model and configuration of your unit is also present. Copy the entirety of the FCDATA01 folder located within
your unit model folder from the upgrade kit to a blank SD card. Safely remove the SD card from your computer, and install it into the PLC. This can be done while the unit is on, however in some cases the unit may require power cycling to initialize the newly installed configuration files.

Once complete, verify your configuration by once again navigating to the **SYSTEM INFO** page.
Press and hold the system time located on the top right corner of the HMI screen for approximately 5 seconds, then press the **SYSTEM INFO** button located on the upper left side of the HMI. You should see your Unit Model displayed at the center of the screen.



Example of an Upgraded and provisioned C270 Water Multi Pass Heat Pump.
Your Serial Number will vary.

You may need to re-enter your Unit serial number to proceed. The Serial Number is typically on a sticker on the top right corner of the interior electrical enclosure of the heat pump. Identify your Serial Number sticker, as a code preceded by "**NHP**" with six digits following. If your Serial Number is missing, damaged, or otherwise unreadable, your serial number is also on the SD card files you backed up prior to conducting this upgrade. The Serial Number will be located in the folder:

Your_Original_File_backup_location\FCDATA01\RECIPE\SN.csv

This file may be opened with a text editor and contains your serial number.

If the Serial Number is illegible, and the SN.csv file is not present, please contact support to verify the Unit Serial Number. The Unit Serial Number is reconfigurable locally, but entering an incorrect serial number could create operational and support issues.

Your R513A C-series Heat Pump is now Upgraded.